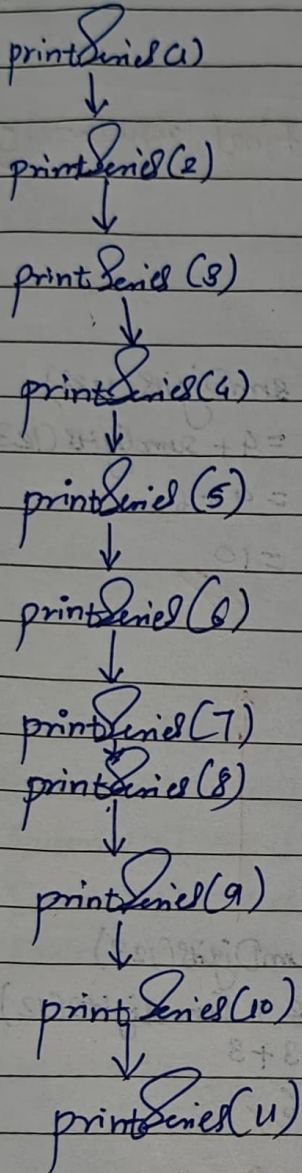


DAY 4

Recursion

DATE		
29	09	25

head recursion

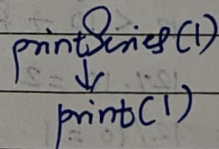
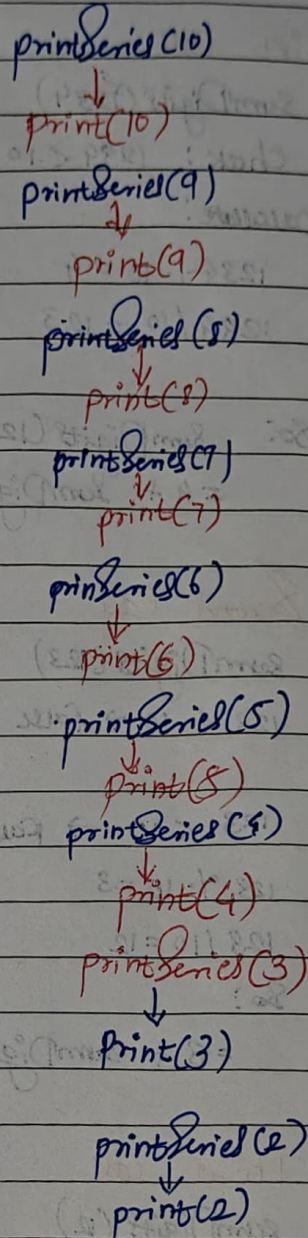


if (n > 10) ✓

|| > 10 return true

printSeries(11) ends

Coming back - printing happens



output:-

10
9
8
7
6
5
4
3
2
1

* Sum of digits Using Recursion

1. First Call

$\text{SumDigits}(1234)$

check: $1234 < 10 \rightarrow \text{false}$

calculate:

$$1234 \% 10 = 4$$

$$1234 / 10 = 123$$

Final result = 10

So: $\text{SumDigits}(1234)$
 $= 4 + \text{SumDigits}(123)$

$\text{SumDigits}(1234)$
 $= 4 + \text{SumDigits}(123)$
 $= 4 + 6$
 $= 10$

2. Second Call

$\text{SumDigits}(123)$

$123 < 10 \rightarrow \text{false}$

$123 < 10 \rightarrow \text{false}$

$$123 \% 10 = 3$$

$$123 / 10 = 12$$

So:
 $= 3 + \text{SumDigits}(12)$

$\text{SumDigits}(123)$
 $= 3 + \text{SumDigits}(12)$
 $= 3 + 3$
 $= 6$

3. Third Call

$\text{SumDigits}(12)$

$12 < 10 \rightarrow \text{false}$

$$12 \% 10 = 2$$

$$12 / 10 = 1$$

So:
 $= 2 + \text{SumDigits}(1)$

$\text{SumDigits}(12)$
 $= 2 + \text{SumDigits}(1)$
 $= 2 + 1$
 $= 3$

4. Forth Call

$\text{SumDigits}(1)$

$1 < 10 \rightarrow \text{true}$

return 1.

recursion returns

* Factorial Using Recursion:

Factorial(5)

- Going down:

Factorial(5)

$$= 5 \times \text{Factorial}(4)$$

$$= 5 \times 4 \times \text{Factorial}(3)$$

$$= 5 \times 4 \times 3 \times \text{Factorial}(2)$$

$$= 5 \times 4 \times 3 \times \text{Factorial}(1)$$

At Factorial(1)

$(n == 0) \vee n == 1$ ✓ return 1

- Coming back:

$$\text{Factorial}(2) = 2 \times 1 = 2$$

$$\text{Factorial}(3) = 3 \times 2 = 6$$

$$\text{Factorial}(4) = 4 \times 6 = 24$$

$$\text{Factorial}(5) = 5 \times 24 = 120$$

Output = 120

Fibonacci Series Using Recursion

Fibonacci(5)

Formula : $\text{fibonacci}(n-1) + \text{fibonacci}(n-2)$

So:

$$\begin{aligned}
 &\text{fib}(5) \\
 &= \text{fib}(4) + \text{fib}(3) \\
 &= (\text{fib}(3) + \text{fib}(2)) + (\text{fib}(2) + \text{fib}(1)) \\
 &= ((\text{fib}(2) + \text{fib}(1)) + (\text{fib}(1) + \text{fib}(0))) + ((\text{fib}(1) + \text{fib}(0)) + 1)
 \end{aligned}$$

Base Case

$$\text{fib}(1) = 1$$

$$\text{fib}(0) = 0$$

Now return:

$$\text{fib}(2) = 1 + 0 = 1$$

$$\text{fib}(3) = 1 + 1 = 2$$

$$\text{fib}(4) = 2 + 1 = 3$$

$$\text{fib}(5) = 3 + 2 = 5$$